

Data Dictionary

Document Metadata

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1. Introduction

This Data Dictionary serves as the authoritative metadata reference and bridge between business terminology and the technical data schemas within the AdventureWorks Sales Domain semantic model.

Rather than listing low-level SQL database types, this document focuses strictly on **business-critical columns** categorized by their modeling **Type** (such as PK, FK, Business Key, Measure, or Attribute) and their **direct physical sources** of our semantic measures. This selective, business-first approach ensures the dictionary remains highly actionable, maintainable, and aligned with our star schema design.

2. Fact Tables

Fact tables store the quantitative, measurable metrics of our business processes.

2.1 FactInternetSales

- **Business Process:** Direct-to-Consumer (B2C) online transactions.
- **Grain:** One row per individual product SKU per sales order line item.

Column Name	Type	Business Description	Analytical & Modeling Usage
SalesOrderNumber	Business Key	Unique identifier for the customer e-commerce order.	Used for drill-through reporting and order-count analysis.
SalesOrderLineNumber	Business Key	Line item identifier within a specific sales order.	Establishes grain uniqueness when combined with

			order number.
ProductKey	FK	Foreign key pointing to the sold product.	Joins to DimProduct for product performance analysis.
CustomerKey	FK	Foreign key mapping to the purchasing customer profile.	Joins to DimCustomer for demographic slicing.
PromotionKey	FK	Foreign key linking any active promotional campaign applied.	Joins to DimPromotion to measure marketing ROI.
CurrencyKey	FK	Foreign key defining the transaction currency.	Joins to DimCurrency for currency conversion mechanisms.
SalesTerritoryKey	FK	Foreign key linking the transactional territory.	Joins to DimSalesTerritory for geographic aggregation.
OrderDateKey	FK	Integer date key representation of order placement.	Active Relationship with DimDate for sales trend analysis.
ShipDateKey	FK	Integer date key representing dispatch from warehouse.	<i>Inactive Relationship</i> with DimDate (activated via DAX for shipping lag).
DueDateKey	FK	Integer date key representing scheduled delivery	<i>Inactive Relationship</i> with DimDate (activated

		date.	via DAX for SLA analysis).
OrderQuantity	Measure	The physical count of units sold within the line item.	Ingested by DAX metric [Total Sales Units].
UnitPrice	Attribute	Unit catalog price set at the moment of checkout.	Used to audit discounts and pricing variances.
ExtendedAmount	Measure	Gross sales value before discounts ($\text{OrderQuantity} \times$).	Baseline calculation audit column.
TotalProductCost	Measure	Standard product production or acquisition cost (COGS).	Ingested by DAX metric [Total Product Cost].
SalesAmount	Measure	Net revenue from the transaction after applying discounts.	Ingested by DAX metric [Internet Sales Revenue].
TaxAmt	Measure	Tax amount charged on the transaction.	Used for net-of-tax and tax margin calculations.
Freight	Measure	Shipping and delivery charges paid by the customer.	Used for logistical margin and fulfillment expense metrics.

2.2 FactResellerSales

- **Business Process:** Business-to-Business (B2B) wholesale distributor transactions.
- **Grain:** One row per individual product SKU per reseller purchase order line item.

Column Name	Type	Business Description	Analytical & Modeling Usage
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SalesOrderNumber	Business Key	Unique identifier for the wholesale purchase order.	Used for corporate order auditing and drill-through.
ProductKey	FK	Foreign key mapping to the distributed product SKU.	Joins to DimProduct for inventory velocity tracking.
ResellerKey	FK	Foreign key mapping to the purchasing corporate client.	Joins to DimReseller for distributor sales tracking.
EmployeeKey	FK	Foreign key identifying the managing sales representative.	Joins to DimEmployee for sales team commission and quota audit.
PromotionKey	FK	Foreign key linking wholesale trade agreements.	Joins to DimPromotion to evaluate B2B trade incentives.
CurrencyKey	FK	Foreign key defining the reporting transaction currency.	Joins to DimCurrency for financial consolidation.
SalesTerritoryKey	FK	Foreign key linking the corporate sales territory.	Joins to DimSalesTerritory for regional market share analysis.
OrderDateKey	FK	Date key representing bulk purchase order date.	Active Relationship with DimDate for timeline trending.
ShipDateKey	FK	Date key representing	<i>Inactive Relationship</i> with

		dispatch to the reseller.	DimDate for lead-time calculation.
DueDateKey	FK	Date key representing delivery commitment date.	<i>Inactive Relationship</i> with DimDate for on-time performance.
OrderQuantity	Measure	Total volume of units purchased by the reseller.	Aggregated within commercial volume KPIs.
UnitPrice	Attribute	Negotiated wholesale price per item.	Monitored to evaluate discount structures.
TotalProductCost	Measure	Total standard cost of producing the bulk product.	Used to determine corporate B2B product margins.
SalesAmount	Measure	Net revenue generated from the reseller order line.	Ingested by DAX metric [Reseller Sales Revenue].
TaxAmt	Measure	Wholesale commercial tax applied.	Tracked for financial compliance auditing.
Freight	Measure	Allocated logistics cost for bulk distribution.	Analyzed to assess bulk distribution delivery costs.

2.3 FactSalesQuota

- **Business Process:** Representative and regional performance planning (Targets).
- **Grain:** One record per sales representative per calendar quarter.

Column Name	Type	Business Description	Analytical & Modeling Usage
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SalesQuotaKey	PK	Surrogate key identifying the quota entry.	Unique identifier for table maintenance.
EmployeeKey	FK	Foreign key assigning target to a sales representative.	Joins to DimEmployee to calculate personal targets.
DateKey	FK	Foreign key assigning target to a specific date period.	Joins to DimDate to identify target calendar quarters.
SalesAmountQuota	Measure	The dollar revenue milestone established as a target.	Ingested by DAX metric [Sales Quota].

3. Dimension Tables

Dimension tables provide descriptive context and hierarchy filters for qualitative slicing.

5.1 DimDate

- **Purpose:** Standardizes reporting periods across calendar and fiscal years.
- **Keys:** DateKey (PK)

Column Name	Type	Attribute Description	Analytical Filter Usage
DateKey	PK	Unique surrogate key formatted as YYYYMMDD.	Joins directly to active and inactive keys in all facts.
FullDateAlternateKey	Attribute	Standard SQL date representation.	Used for continuous time-series charts and date tooltips.
DayNumberOfWeek	Attribute	Numerical day index (e.g., 1 for	Used for sorting day names and

		Sunday, 7 for Saturday).	analyzing weekend/weekday trends.
EnglishDayNameOf Week	Attribute	Textual day name.	Slicing and filtering sales by daily cycle.
MonthNumberOfYear	Attribute	Numerical month index (1 to 12).	Crucial sorting key for Month Name attributes.
EnglishMonthName	Attribute	Textual month name.	Standard monthly comparative reporting (e.g., Jan, Feb).
CalendarQuarter	Attribute	Standard calendar quarter (1 to 4).	Medium-term aggregation filtering.
CalendarYear	Attribute	Standard calendar year (e.g., 2026).	Top-level annual trend filtering.
FiscalQuarter	Attribute	AdventureWorks fiscal planning quarter.	Standardizes internal financial reviews.
FiscalYear	Attribute	AdventureWorks fiscal planning year.	Standardizes annual corporate budget allocations.

5.2 DimProduct

- **Purpose:** Centralized catalog containing product SKUs, subcategories, and categories.
- **Keys:** ProductKey (PK)

Column Name	Type	Attribute Description	Analytical Filter Usage
ProductKey	PK	Unique surrogate key for each product record.	Primary join to Fact tables.

EnglishProductName	Attribute	Descriptive name of the specific product SKU.	Deepest level of product report tables and search filters.
Color	Attribute	Surface color variant of the SKU.	Slicing product preferences (e.g., black frame sales).
Size	Attribute	Physical dimensions / frame size of the SKU.	Customer sizing profile analytics.
ModelName	Attribute	Unified design family representing the item.	Aggregated product performance without SKU size noise.
ProductSubcategoryName	Attribute	Medium-tier category mapping (denormalized).	Dynamic subcategory grouping and drilling.
ProductCategoryName	Attribute	High-level commercial grouping (denormalized).	Top-tier product portfolio filters (e.g., Bikes, Clothing).
Status	Attribute	Operational lifecycle flag (e.g., "Current", "Null").	Filter to isolate active items from obsolete product lines.

Note: ProductSubcategoryName and ProductCategoryName are denormalized from DimProductSubcategory and DimProductCategory respectively into the primary dimension to construct a strict star schema.

5.3 DimCustomer

- **Purpose:** Demographic and profile mapping of direct B2C internet sales clients.
- **Keys:** CustomerKey (PK), GeographyKey (FK)

Column Name	Type	Attribute Description	Analytical Filter Usage
CustomerKey	PK	Unique surrogate key for individual B2C buyers.	Primary join to consumer sales facts.
FirstName / LastName	Attribute	Customer's formal identity records.	Concatenated into Full Name for CRM lookup.
Gender	Attribute	Gender categorization ("M" / "F").	Demographical buying behavior segmentation.
MaritalStatus	Attribute	Marital status ("S" / "M").	Consumer profile clustering.
YearlyIncome	Attribute	Estimated gross annual household earnings.	Grouped into categorical income brackets (Affluence analysis).
EnglishEducation	Attribute	Educational background level.	Socioeconomic lifestyle segmentation.
EnglishOccupation	Attribute	Active occupational sector.	Slicing product selections by professional profiles.
GeographyKey	FK	Key referencing geography database table.	Joins to DimGeography to localize customer locations.

5.4 DimReseller

- **Purpose:** Corporate distributor and retail partner profiles.
- **Keys:** ResellerKey (PK), GeographyKey (FK)

Column Name	Type	Attribute Description	Analytical Filter Usage
ResellerKey	PK	Unique surrogate key for business accounts.	Primary join to wholesale facts.
ResellerName	Attribute	Legal trade name of the corporate partner.	Dynamic lists of corporate customer sales ranks.
BusinessType	Attribute	Retail category classification.	Slicing B2B sales (e.g., Specialty Stores, Warehouses).
ProductLine	Attribute	Primary product specialization segment.	Identifies partner sales profiles (Road bikes vs. Touring).
NumberEmployees	Attribute	Scalability metric of the reseller organization.	Correlates reseller sizes with bulk purchase volumes.
AnnualSales	Attribute	Reseller's declared gross annual market revenue.	Evaluates distributor tiers against actual purchase patterns.
GeographyKey	FK	Geographic location key.	Joins to DimGeography for wholesale region mapping.

5.5 DimEmployee

- **Purpose:** Organizational team records and representative classifications.
- **Keys:** EmployeeKey (PK)

Column Name	Type	Attribute Description	Analytical Filter Usage
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EmployeeKey	PK	Unique surrogate key for the employee.	Direct join to Reseller and Quota fact tables.
FirstName / LastName	Attribute	Employee identity records.	Concatenated for performance scorecards.
Title	Attribute	Active corporate functional title.	Isolates "Sales Representatives" from other corporate roles.
SalesPersonFlag	Attribute	Boolean flag isolating sales active personnel (1 =).	Filter to ensure only target-relevant staff are audited.
ParentEmployeeKey	FK	Self-referencing column linking to direct supervisor.	Modeled in DAX using parent-child hierarchies for team aggregation.

5.6 DimPromotion

- **Purpose:** Tracks discounts and specific pricing campaigns.
- **Keys:** PromotionKey (PK)

Column Name	Type	Attribute Description	Analytical Filter Usage
PromotionKey	PK	Unique surrogate key for promotional campaigns.	Direct join to Internet and Reseller sales facts.
EnglishPromotionName	Attribute	Public or internal title of the discount.	Segments transaction sales by active campaign naming.
DiscountPct	Attribute	Rate of pricing deduction applied	Evaluates price elasticities and

		to line item.	revenue impacts.
EnglishPromotionType	Attribute	Promotion format (e.g., "Volume Discount", "No Discount").	Evaluates corporate promo category performance.

5.7 DimSalesTerritory

- **Purpose:** Corporate sales boundary classifications.
- **Keys:** SalesTerritoryKey (PK)

Column Name	Type	Attribute Description	Analytical Filter Usage
SalesTerritoryKey	PK	Unique surrogate key representing territory boundaries.	Direct join to Sales fact tables.
SalesTerritoryRegion	Attribute	Commercial region names (e.g., Northwest, Southwest).	Direct filtering on localized territory structures.
SalesTerritoryCountry	Attribute	Regional country boundaries (e.g., United States).	Regional and global reporting segmentation.
SalesTerritoryGroup	Attribute	Multi-national groupings (e.g., North America, Europe).	Corporate executive-level spatial filters.

5.8 DimGeography

- **Purpose:** Geospatial tracking of physical customer and reseller properties.
- **Keys:** GeographyKey (PK)

Column Name	Type	Attribute Description	Analytical Filter Usage
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GeographyKey	PK	Unique surrogate key for location attributes.	Direct link to B2C and B2B parent dimensions.
City	Attribute	Municipality name.	Deepest level of demographic spatial maps.
StateProvinceName	Attribute	State or regional administration names.	Regional cluster metrics.
EnglishCountryRegionName	Attribute	Standardized country name.	Top-tier comparative demographic maps.

4. Analytical Hierarchies

To support structured Drill-Down analysis within report visuals, the following hierarchies must be explicitly defined in the semantic model:

4.1 Product Hierarchy

Enables users to drill from top-level macro portfolios down to specific transactional item models.

Product Category Name $\xrightarrow{\text{Drill Down}}$ Product Subcategory Name $\xrightarrow{\text{Drill Down}}$ English Product Name

4.2 Geography Hierarchy

Allows geospatial sales tracking, comparing regions from high-level country groups down to individual customer cities.

Sales Territory Group $\xrightarrow{\text{Drill Down}}$ English Country Region Name $\xrightarrow{\text{Drill Down}}$ State Province Name

5. Semantic KPI to Physical Data Source Mapping

This matrix maps core logical business KPIs directly to their physical table columns and initial aggregation rules:

KPI ID	KPI Name	Primary Physical Source Table	Source DB Column
FIN-01	Total Revenue	FactInternetSales & FactResellerSales	SalesAmount
FIN-02	Total Product Cost	FactInternetSales & FactResellerSales	TotalProductCost
FIN-03	Total Gross Profit	FactInternetSales & FactResellerSales	<i>Derived</i>
FIN-04	Gross Profit Margin %	FactInternetSales & FactResellerSales	<i>Derived</i>
FIN-05	Total Sales Units	FactInternetSales & FactResellerSales	OrderQuantity
SAL-01	Internet Sales Revenue	FactInternetSales	SalesAmount
SAL-02	Reseller Sales Revenue	FactResellerSales	SalesAmount
SAL-04	Sales Quota	FactSalesQuota	SalesAmountQuota
SAL-05	Quota Achievement %	FactResellerSales & FactSalesQuota	<i>Derived</i>